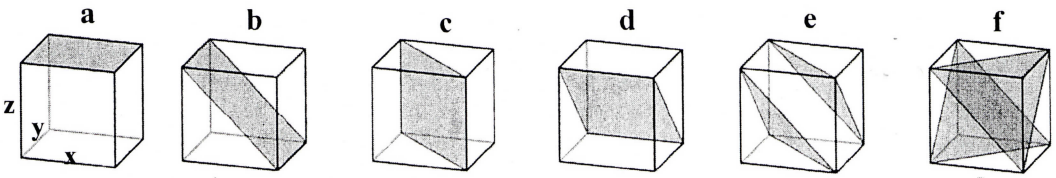
Inorganic CUME on Materials Chemistry
October 26th, 2018

Answer any five questions below. Only five will be graded.

1. (20 points) Label the Miller Indices for each shaded plane below.



2. (20 points) There are many different approaches for materials synthesis. Examples are (a) solid state synthesis, (b)sol-gel methods, (c) hydrothermal synthesis, (d) chemical vapor deposition, (e) combustion synthesis and (f) electrochemical synthesis. Discuss how you can prepare ZnO from 4 of those methods.

3. (20 points) Iron(II) forms a complex in hemoglobin. For each of the two octahedral complex ions $[Fe(H_2O)_6]^{2+}$ and $[Fe(CN)_6]^{4-}$, draw an energy diagram showing orbital splitting, predict the number of unpaired electrons, and identify the ion as low spin or high spin.

4. (20 points) Use partial orbital diagrams to describe how mixing of the atomic orbitals of the central atom(s) leads to hybrid orbitals in each of the following: (a) Methanol, CH_3OH; (b)Sulfur tetrafluoride, SF_4

5. (20 points) Explain what is (a) Schottky defect, (b) Frenkel defect and (c) Stacking Fault. What are the differences between intrinsic and extrinsic defects?

6. Explain why crystalline solids are generally more defective as a result of increasing temperature? Explain why magnetic materials that are to be used in information storage should have either square or rectangular shaped hysteresis loops.

7. (20 points) What is cyclic voltammetry (CV). Draw a typical cyclic voltammogram using Current (I) vs. Potential (V). How can CV be used to determine whether it is Faraday or non-Faraday currents? How to use CV to determine the reversibility of a redox reaction and the stability of reaction products?

8. (20 points) A copper substrate has been covered by a thin film of inorganic nanocrystals. List at least three different characterization techniques that can be combined to study the properties of coating layer including the chemical composition, structure, surface area, thickness and uniformity.

9. (20 points) Give the coordination number and charge of the central metal ion in (a) $Na_2[Zn(OH)_4]$; (b) $K[Co(H_2O)_2(C_2O_4)_2]$; (c) $[Ru(H_2O)_2(NH_3)_2Cl_2]Br$.

10. (20 points) Zn metal and many other first row metals readily dissolve in acid and give hydrogen gas. Copper, however, is essentially nonreactive. Rationalize this observation.